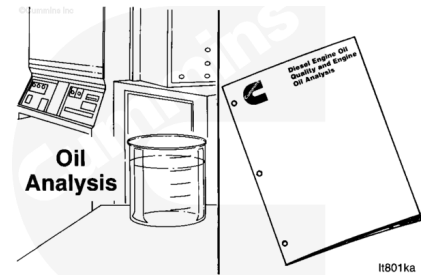


Trainings ▾

General Information

A used oil analysis can help diagnose internal damage and determine if it was caused by one of the following:

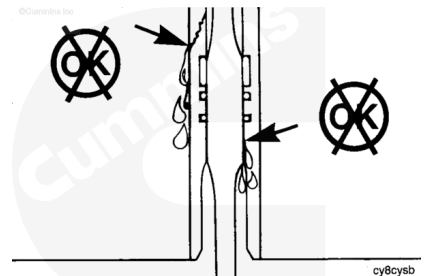
- Oil diluted with coolant
- Oil diluted with fuel.



Pressure Test

Coolant-Diluted Lubricating Oil

For lubricating oil that is diluted with coolant, Perform the cooling system pressure test. Refer to Procedure 008-018 ([/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html](http://qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)).



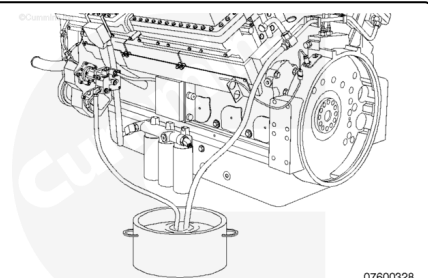
Fluorescent Dye Tracer

Lubricating Oil Dilution

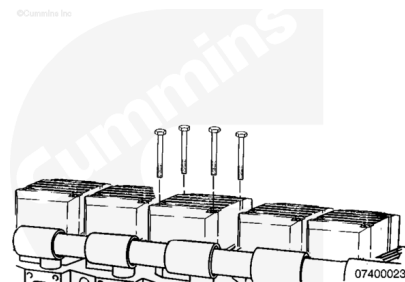
Perform this test on oil that is diluted with fuel.

This test is **not** effective on a cold engine, less than 21°C [70°F] coolant temperature, or with a loose overhead setting.

Install an isolated fuel supply tank to the inlet and drain lines. Add fluorescent tracer, Part Number 3376891, to the fuel supply tank.



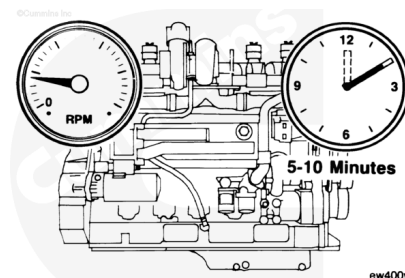
Remove the valve cover mounting capscrews, but do **not** remove the valve covers.



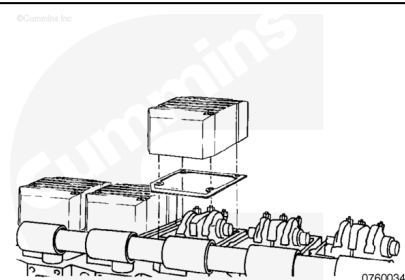
Start the engine and operate it at high idle for 30 seconds.

Run the engine at low idle.

Complete the next step within 5 minutes.

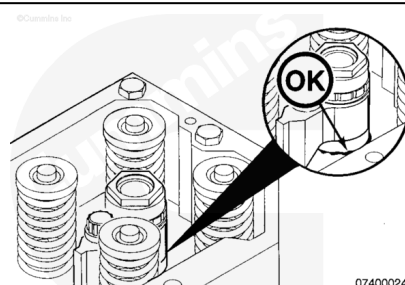


While the engine is operating at low idle, remove the valve covers and gaskets. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>)

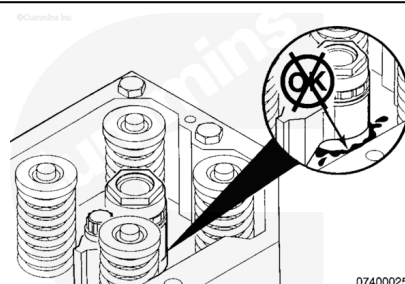


Injectors normally have a small amount of fuel leakage. Fuel will be a yellow color.

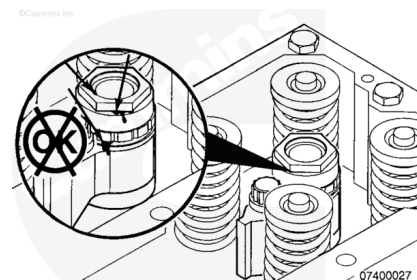
Use a black light to find fuel leaks from inside or around the injector.



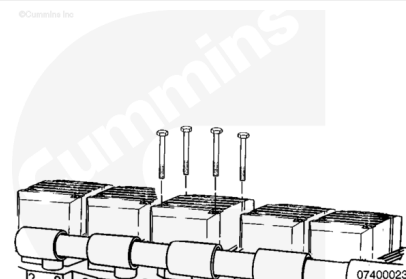
If there is excess leakage around the outside of the injector, remove the injector and replace all four o-rings. Refer to Procedure 006-026 (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>).



If there is excess leakage from inside the injector, replace the injector. Refer to Procedure 006-026 (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>).



Install the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).

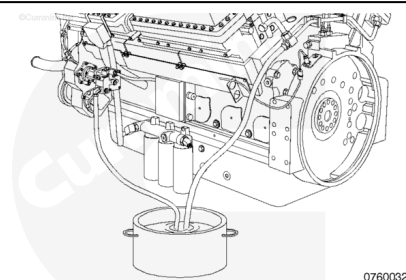


QSK60 Marine Applications

Perform this test on oil that is diluted with fuel.

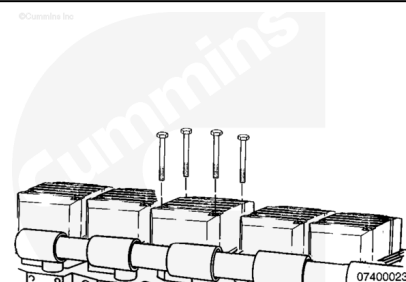
This test is **not** effective on a cold engine, less than 21°C [70°F] coolant temperature, or with a loose overhead setting.

Install an isolated fuel supply tank to the inlet and drain lines. Add fluorescent tracer, Part Number 3376891, to the fuel supply tank.



Remove the aftercooler heat shields. Refer to Procedure 010-129 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>).

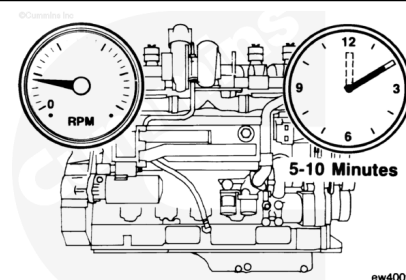
Remove the valve cover mounting capscrews, but do **not** remove the valve covers.



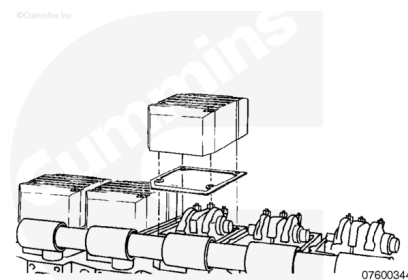
Start the engine and operate it at high idle for 30 seconds.

Run the engine at low idle.

Complete the next step within 5 minutes.

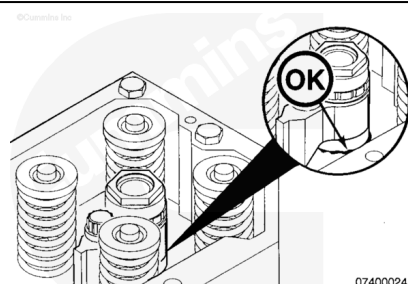


While the engine is operating at low idle, remove the valve covers and gaskets. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>)

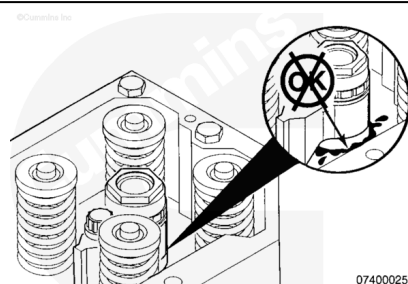


Injectors normally have a small amount of fuel leakage. Fuel will be a yellow color.

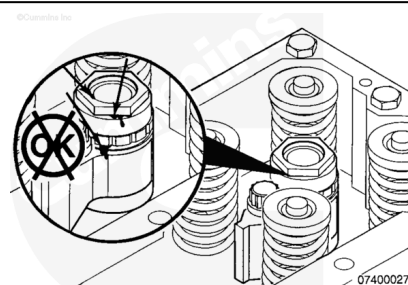
Use a black light to find fuel leaks from inside or around the injector.



If there is excess leakage around the outside of the injector, remove the injector and replace all four o-rings. Refer to Procedure 006-026 (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>).



If there is excess leakage from inside the injector, replace the injector. Refer to Procedure 006-026 (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>).



Install the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).

Install the aftercooler heat shields. Refer to Procedure 010-129 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>).

